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Geny F. Lapiña

Department of Agricultural and Applied Economics

Aileen V. Lapitan

Institute for Governance and Rural Development

Samantha Joyce P. Manalastas

Department of Agricultural and Applied Economics

Charis Mae T. Neric

Department of Agricultural and Applied Economics



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Geny F. Lapiña¹, Aileen V. Lapitan², Samantha J.P. Manalastas¹, and Charis Mae T. Neric¹

¹Department of Agricultural and Applied Economics, College of Economics and Management, University of the Philippines Los Baños, College, Los Baños, Laguna, Philippines

²College of Public Affairs and Development, University of the Philippines Los Baños, College, Los Baños, Laguna, Philippines

Abstract

Global challenges such as hunger, malnutrition, and environmental degradation have intensified the need to transform food systems, making it a critical agenda for sustainable development, particularly in relation to the food and agriculture sectors. This transformation necessitates a radical shift, not only in increasing food supply but also in integrating food security with healthy diets, improving livelihoods, and fostering environmental sustainability. However, the readiness of the Philippines' food system for such transformation remains a challenge, despite various government policies and programs aimed at enhancing food production and nutrition. This paper examines the policies and governance frameworks in the Philippines to assess their role in preparing the country's food systems for transformation. It explores potential overlaps and gaps in these policies and provides insights for future research and policy considerations to achieve a cohesive strategy for food systems transformation in the country.

Keywords: food and nutrition systems transformation, food policy, governance

INTRODUCTION

In the face of the global challenges of hunger, malnutrition, and environmental degradation, transforming food systems has become the new development agenda for the food and agriculture sector. The food system encompasses all actors and their interlinked activities along the value chain--- from input supply, production, aggregation, processing, distribution, consumption, down to the disposal of food commodities in the agriculture, forestry, or fisheries sectors ([FAO, n.d.](#)). A growing population depends on food systems for its survival, at the expense of the sustainability of natural ecosystems. The costly implications of this balancing act are the motivation for a global initiative toward food system transformation. Case in point, the World Bank's [Food Systems 2030](#) supports food systems transformation toward the triple target of human, planetary, and economic health.

Food systems transformation aligns with the United Nations' Sustainable Development Goals ([SDGs](#)), adopted in 2015, which aim for sustainable food systems worldwide by 2030. SDG2, i.e., ending hunger, achieving food security, improving nutrition, and promoting sustainable agriculture, lists food system transformation alongside imperatives to address entrenched inequalities, invest in sustainable agricultural practices, and mitigate the impact of shocks. The kind of transformation being referred to is characterized by an extent of radicality in terms of not merely targeting an increase in food supply, but also bundling food security with healthy diets, with an eye toward enhancing livelihoods along the food value chain and fostering environmentally beneficial activities (Dengerink, 2022). To achieve this, interconnected actions between local, national, regional, and global levels are required (Nguyen, 2018).

Food security in the Philippines, despite sustained efforts by the government, has had a patchy record of success. The readiness of the country's food system for transformation is crucial to addressing SDG2 for zero hunger. Several policies and programs in support of this goal have been promulgated and implemented by the Philippine government. However, it has yet to be established how the country's policy and governance landscape provides a cohesive environment to prepare communities for a transformation in food systems.

In this paper, we examine the policies and governance systems deemed to foster the readiness of Philippine food systems for transformation. In particular, we aim to identify overlaps and gaps. Finally, we share some insights for consideration in research and policy.

METHODOLOGY

This paper conducts a scan of existing policies and programs that relate to food systems thinking. A thematic analysis of policy documents, government publications, and gray literature will be conducted. A meta-analysis of published literature on food systems transformation will also be done to supplement the analysis of any policy gaps.

A basic text analysis where keywords on themes will be identified using NVivo. This will give a cursory look but also generate some insights. An analytical framework for understanding the Philippines' policy readiness on food system transformation based on the Global Alliance for the Future Food will be used as a guide (2021). Moreover, a qualitative evaluation of implementation (or non-implementation) and consequences (multi and interdisciplinary lens) for food systems transformation vis a vis key policy will be conducted.

REVIEW OF RELATED LITERATURE

Food Systems Framework

A food system, as stated by the Food and Agriculture Organization of the United Nations, consists of different sub-systems such as food, waste management, and input supply. These subsystems interact with key systems such as energy, health, and trade. It encompasses all actors and interlinked value-adding activities involved in agriculture, and parts of the broader economic, societal, and natural environments (2018). A food system is considered sustainable if it delivers food security and nutrition for all without compromising the economic, social, and environmental bases of the future generation (IFAD, n.d.). FAO further explains that a food system is sustainable if it is profitable throughout, it has broad-based benefits for society, and it has a positive or neutral impact on the environment.

As emphasized by FAO (2018), a sustainable food system should reflect qualities that align with the six dimensions of food security. It must be productive and prosperous to guarantee sufficient food availability, equitable and inclusive to ensure access to food and livelihoods for all, respectful and empowering to provide agency for people and groups to make choices and shape the system, resilient to maintain stability during shocks and crises, regenerative to ensure sustainability across all dimensions, and healthy and nutritious to support nutrient uptake and utilization. When a food system has these qualities in an interconnected manner, it is highly likely to meet the goals of SDG2. It would also ultimately support the realization of the right to food.

Food Systems Transformation

Food systems transformation is recognized as the solution to achieving zero hunger, addressing the climate crisis, and preventing biodiversity loss (Cooper, 2023). In food systems transformation, it not only targets an increase in food supply but also healthy diets and fair incentives on the food value chain. It should also be sustainable and environmentally friendly.

Environmental damage, food insecurity and malnutrition, public health problems, and food loss and waste are considered to be the challenges that must be tackled by transforming food systems. Due to rapid population growth, poverty, food insecurity, malnutrition, climate change, land, water, and biodiversity degradation, and food loss and waste, the transformation of food systems must be taken into consideration. Food systems must ensure that health, inclusion, sustainability, resilience, climate, and the environment are their top priorities (Laderchi et al., 2024). A thorough understanding of how food systems function is vital in

assessing these systems and how they should be developed while considering minimizing the negative impacts and maximizing their positive contributions (Nguyen, 2018).

To commit to the SDGs by 2030, Nguyen (2018) states that global food systems should be reshaped to be more productive, environmentally sustainable, resilient, able to deliver healthy food, and more inclusive of the marginalized and the poor. The Food System Economics Commission in their publication entitled “The Economics of the Food System Transformation” reported that for food systems transformation to occur, there should be a shift towards healthy food consumption, a reset on the government support in agriculture for economic incentives in the agricultural and food markets, improvement in research and innovation to increase labor productivity and workers’ livelihood opportunities, and scaling up of safety nets to maintain affordable food for the marginalized (Laderchi et al., 2024).

The Global Alliance for Improved Nutrition launched the Initiative on Climate Action and Nutrition (I-CAN) in 2022 reports addresses the climate and nutrition challenges and their linkages. A baseline across 20 indicators was grouped into four pillars to analyze the extent of integration between climate and nutrition action. The 2023 baseline report showed gaps at the implementation, action, and support levels, at the data and knowledge levels, and at the policy level. With this, it was highly suggested that these baselines be used by countries in tracking their indicators, which will help the government in assessing progress and identifying areas that need support (2024).

However, it is a question of how we will transform the food systems. We then consider the United Nations Food Summit and the shifting of planetary healthy diets, which promote less meat and more plant-based diets. The United Nations Food Summit considered action tracks to support the transformation of food systems to achieve the goals of the SDGs. The five action tracks are expected to focus on their specific goals but must be understood to be mutually supporting and interconnected to ensure holistic, coordinated action. The Action Tracks emphasize a multi-stakeholder approach and are structured to support the broader SDG framework, with a focus on overcoming barriers and facilitating inclusive actions. They are designed to reflect the complexity of food systems and are structured around five main areas: ensuring access to nutritious food for all, shifting to sustainable consumption patterns, boosting nature-positive production, advancing equitable livelihoods, and building resilience to vulnerabilities and shocks.

The transformation of global food systems is a critical step toward addressing some of the most pressing challenges of our time, including hunger, climate change, and biodiversity loss. Achieving this transformation requires a multi-faceted approach that integrates initiatives

like the United Nations Food Summit's action tracks or the adoption of planetary health diets. By shifting towards more sustainable and inclusive food systems, and by supporting policies that prioritize health, equity, and environmental sustainability, a more resilient and equitable global food system can be created. While challenges remain, coordinated global efforts can drive meaningful progress toward a sustainable future for all.

Philippine Food and Nutrition System

The UN FAO State of Agriculture and Food highlighted the economic impact of food-related issues in the Philippines, showing a value of 95 billion USD in 2021 Purchasing Power Parity (PPP) for total hidden food costs. Reports also showed significantly high health costs in the Philippines. Specifically, the country showed a low consumption of plant-based whole foods, with a value of 43 billion USD in 2021 PPP. In contrast, the high consumption of processed foods and additives showed a value of 21 billion USD, contributing significantly to the overall health burden.

In the context of the food chain in the Philippines, food security is often equated with rice security. This implies that most of the attention from the public and policymakers is focused on rice. Within the food chain, it is commonly observed that only the production and consumer stages receive significant attention. The high allocation of the rice budget by the Department of Agriculture reflects the limited support for diverse and nutritious crops on the production side. On the consumer side, there is emphasis on food prices, healthy diets, and the increasing issues of food insecurity and non-communicable diseases. However, the mid-stages—specifically post-harvest, processing, distribution, and retail—receive less focus, which affects the overall efficiency of food production and consumption. This neglect may lead to limited and unsustainable access to food.

State of Food Security and Food System in the Philippines

Based on the 2021 Global Food Security Index from The Economist Intelligence Unit, the Philippines ranked 64th out of 113 countries in terms of the four dimensions of food security (Galang, 2022). Despite these challenges, the Philippines has implemented numerous policies to tackle food, nutrition, and hunger challenges. However, these efforts often lack coordination and fail to consider the interconnected nature of food systems, leading to inefficiencies and socioeconomic disparities (Golloso-Gubat et al., 2024). Lapiña and Catelo highlight the significant research gap and knowledge about food security in the Philippines. Research on food security is generally dispersed, and a better understanding of the demand side of food

security, as well as unified research for development framework and action plan, is strongly needed (2024).

Policy and Governance in Food Systems of the Philippines

Active policy efforts are crucial to managing the interconnected elements of food systems. In the food systems diagram, policy and governance directly and significantly affect various components and influence how each component affects the other. We examined the five major government departments that affect our food systems: the Department of Agriculture, the Department of Health, the Department of Trade and Industry, the Department of Science and Technology, and the Department of Environment and Natural Resources.

Department of Agriculture

The Department of Agriculture plays a pivotal role in influencing food supply chains through policies and programs like the Rice Tariffication Law (RTL), the Agriculture and Fisheries Modernization Act (AFMA), and the National Agriculture and Fisheries Modernization and Industrialization Plan (NAFMIP). RTL aims to provide cheap, high-quality, and accessible rice to farmers. Low prices due to tariffication are considered to significantly improve societal welfare, as rice, the nation's staple food, accounts for 20% of a poor household's consumption (PhilRice, 2019). AFMA aims to modernize the agriculture and fisheries sectors, enhance profits and incomes, ensure a stable food supply, and improve the quality of life across all sectors. AFMA also strives to provide social and economic adjustment measures that increase productivity and improve market efficiency while ensuring the protection and preservation of the environment and equity for small farmers and fisherfolk. Government agencies such as the Department of Trade and Industry, the Department of Science and Technology, and the Department of Environment and Natural Resources are tasked to coordinate their investment programs and activities to complement the Department of Agriculture's implementation of AFMA (1997). Furthermore, the National Agriculture and Fisheries Modernization and Industrialization Plan (NAFMIP) is considered a directional plan that aims to be more policy and strategy-oriented, though not as thorough as the previous AFMP iterations. Private and public stakeholders are encouraged to take coordinated, cohesive, and determined actions that will be achieved with minimal impact on the environment and without sacrificing the nutrition and health of consumers (Department of Agriculture, 2022). While these policies aim to enhance agricultural productivity and sustainability, their implementation often faces challenges such as inadequate coordination with other sectors and insufficient support for smallholder farmers (PhilRice, 2019).

Department of Health and Department of Science and Technology

The Department of Health directly affects the nutrition and health outcomes of consumers. The Department of Science and Technology, specifically under the Food and Nutrition Research Institute, also shares a common objective with the DOH. Programs and policies under DOH are the National Nutrition Program, the PPAN, and the Food Fortification Act of 2000. The National Nutrition Program and PPAN's main objective coincides with DOST-FNRI's National Nutrition Survey and Nutritional Guidelines and Dietary Recommendations programs. These target the nutritional status of Filipinos, along with addressing the prevalent malnutrition in the country.

DOH's Food Fortification Act of 2000 mandates the fortification of certain foods, while DOST has Food Fortification Programs that involve research and development of fortified foods to address micronutrient deficiencies. Under Sec 9 of the Republic Act No. 8976, "The Philippine Food Fortification Act of 2000," it states that DTI, DOST, and LBP are the government agencies that shall support the implementation of this act through their respective programs. The Department of Trade and Industry (DTI) is hereby required to assist and support affected manufacturers in upgrading their technologies by helping them through financial assistance. The Department of Science and Technology (DOST) will create and execute extensive programs to acquire, design, and produce machines and technologies for the manufacturers (2000).

Department of Trade and Industry

The Department of Trade and Industry launched the Manufacturing Resurgence Program (MRP) aimed at strengthening agriculture-based manufacturing industries to create jobs and support smallholder farmers and agricultural cooperatives through product development, value addition, and integration with large enterprises for marketing and financing. It also aims to craft and implement projects that ensure the sustainable and inclusive growth of the manufacturing industry. The MRP was implemented with the coordination of key government agencies such as the Department of Science and Technology, the Department of Agriculture, and the Department of Energy. The MRP has significantly improved the ease of doing business across various industries while also addressing competitiveness challenges in specific sectors (Board of Investments, n.d.).

Department of Environment and Natural Resources

The Department of Environment and Natural Resources is responsible for the conservation of the country's natural resources. They ensure the proper use of natural resources

and protect the environment with the goal of sustainable development. Policies/programs such as the National Greening Program (NGP) and the Philippine Clean Water Act are just a few of the many policies/programs they have to ensure that the environment is safe and well-maintained. The National Greening Program aims to reduce poverty, promote food security, environmental stability, and biodiversity conservation, and enhance climate change mitigation and adaptation. Along with DA and the Department of Agrarian Reform, NGP strives to consolidate all greening efforts, grant incentives from agroforestry plantations, and ensure the appropriate use of science and technology (DENR, n.d.). The Clean Water Act then aims to protect and manage water resources to ensure clean and safe water for various uses, including agriculture. It also aims to address the degradation of our water bodies and mandates the bureau to formulate and implement policies and strengthen partnerships with stakeholders to improve water quality (DENR, 2024).

RESULTS AND DISCUSSION

Textual Analysis using NVIVO

Table 1. List of associated words in livelihood, nutrition and health, climate change and environment

	ASSOCIATED WORDS
Livelihood	livelihood farm income
Nutrition and Health	food safety nutrition and health health and nutrition non-communicable diseases nutrition health food safety healthy nutrition NCDs
Climate Change/Environment	climate change environmental sustainability climate resilience climate adaptation

Livelihood, nutrition and health, and climate change and environment are the main criteria or key themes used in assessing the Philippine key policies on agriculture and food, nutrition and health, climate change and environment, and national development. Table 1 shows the text analysis based on words associated with livelihood, nutrition and health, and climate change in Philippine policies. For livelihood, words livelihood and farm income were used. For nutrition and health, the associated words used are food safety, nutrition and health, health and nutrition, non-communicable diseases, nutrition, health, health nutrition, and NCDs. As for climate change and environment, words used are climate change, environmental sustainability, climate resilience, and climate adaptation. These key themes are essential in

assessing the extent of the relation of food security in the policy documents and how often they were tackled.

Table 2. Textual Analysis of Livelihood, Nutrition and Health, and Climate Change and Environment in Philippine Key Policies

PHILIPPINE KEY POLICIES	LIVELIHOOD	NUTRITION AND HEALTH	CLIMATE CHANGE/ ENVIRONMENT
Agriculture and Food			
TOTAL	65	41	143
Nutrition and Health			
TOTAL	14	270	1
Climate Change and Environment			
TOTAL	29	0	630
National Development			
PDP	41	22	133
NDC			99
TOTAL	41	22	232

Based on Table 1, values presented in Table 2 show the word frequency of each key theme in Philippine policy documents. Results showed that policy documents on Agriculture and food seem to be more “holistic” and consider dimensions of livelihood, nutrition and health, and climate change and environment. Meanwhile, climate change and environment, as well as nutrition and health, tend to specialize more on their aspects and focus less on other key themes. PDP or Philippine Development is considered a national development policy that targets all sectors of the Philippines. Hence, it can be noted in the table above that it seemingly considers all key themes, but with emphasis on climate change and the environment.

Table 3. Sub-thematic analysis of Livelihood, Nutrition and Health, Climate Change and Environment in Philippine Key Policies

PHILIPPINE KEY POLICIES	LIVELIHOOD	NUTRITION AND HEALTH		CLIMATE CHANGE AND ENVIRONMENT			
	livelihood	food security	food sovereignty	sustainability	natural resources	renewable resources	resilience
Agriculture and Food							
TOTAL	55	78	5	2	37	2	52
Nutrition and Health							
TOTAL	13	36	0	1	7	0	1
Climate Change and Environment							
TOTAL	29	40	0	0	54	0	171
National Development							
PDP	38	46		2	45	1	113

Table 3 shows a sub-thematic analysis of the key themes in Philippine Policies. The sub-themes were identified based on the word frequency results in Table 2. The most commonly found words in each key theme were used and analyzed directly in Philippine policy documents. Livelihood was used for key themes livelihood, food security, and food sovereignty was used for Nutrition and Health, and sustainability, natural resources, renewable resources, and resilience were the sub-themes used for Climate Change and Environment. Results showed that Philippine Policy documents on Agriculture and Food tackle more on all the sub-themes, while Nutrition and Health show very few word frequency on sub-themes relating to livelihood and climate change and environment. It could also be noted that sub-themes “food sovereignty,” “sustainability,” and “renewable resources” are only partly mentioned in agriculture and food policies and have a very limited scope in all key policies.

Table 4, on the other hand, shows the frequency of departmental engagements in Philippine policy documents. The specified government agencies were considered to be the major departments that affect the Philippine food security. Results showed that the Department of Agriculture is often tackled in policies related to agriculture and food, Department of Health in policies related to health and nutrition. Meanwhile, it can be seen that the Department of Agriculture, Department of Science and Technology, Department of Environment and Natural Resources, Climate Change Commission, and NEDA are commonly found on policy documents related to the environment and climate change. All government agencies are then tackled in the Philippine Development Plan. Results of this textual analysis showed that the government agencies have specific roles and/or functions in the Philippine policy documents. There is a possibility of interconnected engagement between departments in order to implement the policies.

Table 4. Frequency of Departmental Engagements in Philippine Policies

PHILIPPINE KEY POLICIES	DEPARTMENT OF AGRICULTURE	DEPARTMENT OF HEALTH	DEPARTMENT OF SCIENCE AND TECHNOLOGY	DEPARTMENT OF ENVIRONMENT AND NATURAL RESOURCES	CLIMATE CHANGE COMMISSION	NEDA	DTI
Agriculture and Food							
TOTAL	677	27	67	34	0	184	75
Nutrition and Health							
TOTAL	211	542	162	14	0	25	34
Climate Change and Environment							
TOTAL	100	42	102	189	176	133	60
National Development PDP							
PDP	124	142	138	105	18	236	306

Table 5. Textual Analysis of Transformation and Systems Thinking in Philippine Key Policies

PHILIPPINE KEY POLICIES	TRANSFORMATION	SYSTEMS THINKING
Agriculture and Food		
NAFMIP	206	34
AFMA	2	
Nutrition and Health		
PPAN	3	
Climate Change and Environment		
NAP	10	2
NDC	3	
RA 9003 Solid Waste Act	1	
National Development		
PDP	114	1

Transformation and Systems Thinking are the vital aspects needed in a food system transformation. Table 5 presents the analysis of whether the Philippine policies tackle transformation or systems thinking. Results showed that only agriculture and food, through the National Agriculture and Fisheries Modernization and Industrialization Plan or NAFMIP, is prominent in transformation and food systems thinking. Philippine policies on nutrition and health, and climate change, and environment are less explicit on transformation and systems thinking. Philippine Development Plan or PDP, on the other hand, shows that transformation is often discussed as a development plan.

Table 6. Textual Analysis on the Philippines' policy readiness for food system transformation

PHILIPPINE KEY POLICIES	POLICY COHERENCE WITH PRINCIPLES							INSTITUTIONAL CAPACITY AND GOVERNANCE	RESOURCE AVAILABILITY
	Healthfulness	Inclusion	Equity	Renewability	Diversity	Interconnectedness	Resilience		
Agriculture and Food									
TOTAL	114	35	129	37	37	0	3	417	63
Nutrition and Health									
TOTAL	521	14	53	7	7	1	0	131	6
Climate Change and Environment									
TOTAL	305	32	0	55	79	1	3	284	27
National Development									
PDP	623	95	95	46	26	5	1	332	6

Table 6 shows the results of a textual analysis to assess the degree to which Philippine policy documents reflect key principles, institutional capacity and governance, and resource availability, which are necessary for food system transformation. Results show that healthfulness is strongly emphasized in the four key policy areas, particularly in Nutrition and Health and National Development. Equity and inclusion, however, appear less frequently across all sectors, indicating potential policy gaps in addressing social dimensions. The Philippine policies on Climate Change and Environment show greater attention to diversity and renewability. Institutional capacity and governance are well-represented, while interconnectedness, resilience, and resource availability show relatively low frequency among all Philippine policies.

Budget allocation on Government Agencies

Table 7. Budget allocation of Government Agencies, 2023-2024

RANK	DEPARTMENT	2023 BUDGET (in Billion Pesos)	2024 BUDGET (in Billion Pesos)	PERCENT CHANGE
1	Department of Education	895.2	924.7	3.30%
2	Department of Public Works and Highways	894.2	822.2	-8.05%
3	Department of Health	314.7	306.1	-2.73%
4	Department of the Interior and Local Government	253.2	259.5	2.49%
5	Department of National Defense	203.4	232.2	14.16%
6	Department of Transportation	106.0	214.3	102.17%
7	Department of Social Welfare and Development	199.5	209.9	5.21%
8	Department of Agriculture	173.6	181.4	4.49%
9	The Judiciary	54.9	57.8	5.28%
10	Department of Labor and Employment	47.1	40.5	-14.01%

Note. From the 2024 People's Proposed Budget, Department of Budget and Management

Budget allocation to government agencies is crucial in food systems transformation. Adequate and strategic budgeting is essential for government agencies to implement programs that support sustainable agriculture, improve nutrition outcomes, and address environment and climate change issues. Without sufficient budget for each government sector, policy coherence, inter-agency collaboration, and resource availability remain limited, which may result in fragmented coordination. In the 2024 People's Proposed Budget, the top priority sector is the Department of Education. The Department of Health and the Department of Agriculture rank third and eighth, respectively. It can also be noted that there was a change in the 2024 budget for the Department of Health with a 2.73 percent decrease from year 2023. Meanwhile, the

Department of Agriculture shows a boost in the budget from 2023, with a 4.49 percent increase in the budget.

SUMMARY AND POLICY RECOMMENDATIONS

The role of policy and governance in shaping the food system of the Philippines is crucial in achieving food security, nutrition, and environmental sustainability. As outlined by the United Nations' Sustainable Development Goals (SDGs), transforming global food systems by 2030 requires coordinated efforts across local, national, regional, and global levels (Nguyen, 2018). Despite efforts to address food security and nutrition challenges, the Philippines faces persistent issues such as inadequate policy coordination, limited resources for smallholder farmers, and the impacts of climate change (Galang, 2022; Goloso-Gubat et al., 2024). These challenges often lead to overlapping mandates and inconsistent policy outcomes, affecting the holistic development of the food system.

To address these challenges, enhanced collaboration and synergy among government agencies and multi-stakeholders are crucial. Implementing integrated policy frameworks that consider the interconnected nature of food systems could streamline efforts and maximize impact. Strengthening monitoring and evaluation mechanisms, supported by robust data and research, can ensure the effectiveness and sustainability of policies aimed at improving food security, nutrition outcomes, and environmental sustainability. One can also consider leveraging the four pillars of I-CAN in 2023 to address the linkages between climate and nutrition in the Philippine food system. The indicators under the four pillars can assess the level of integration, ranging from level 1 with no integration to level 4 with strong integration and concrete action plans (GAIN, 2024). FAO's methodologies for measuring, analyzing, and improving food system performance can also be adapted to better address challenges encountered by the Philippine food system (2018). Achieving a sustainable and resilient food system in the Philippines requires proactive policy measures, robust governance structures, and inclusive stakeholder engagement. By identifying the root causes of these problems, policymakers can address these challenges in a more coherent and coordinated way.

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